

Id no:25dce049

Practical 5: Registration Form with Frontend Validation and User-Friendly Error Handling

Problem Definition Create a student registration form with HTML5 input types and JavaScript validation for name, email, mobile number, password, confirm password, course, year, gender, and terms acceptance. Use Regular Expression for validations.

Key Questions / Analysis / Interpretation

- 1. Are correct input types and attributes used?**
- 2. Are validation errors displayed near the relevant fields?**
- 3. Is password strength checked?**
- 4. Is the form accessible using keyboard and screen-reader-friendly labels?**

Key Skills to be Addressed

HTML forms, JavaScript validation, regular expressions, password strength meter, UX error messages

Applications Registration and data-entry systems

Learning Outcome Students will design accessible forms with robust client-side validation.

CO Mapping CO1, CO3

Dataset/Test Data Valid and invalid sample registration inputs

Tools/Technology to be Used

HTML5, JavaScript ES5/ES6, browser dev tools

Total Hours Implementation: 4 hours | Total engagement: 6 hours
Post Laboratory Work
Description

Submit validation test cases with screenshots for valid and invalid inputs.

Evaluation Strategy
Including Viva

Input coverage, validation accuracy, error handling, and code walkthrough.

Advanced/Intermediate
Extension

Intermediate: Add real-time validation on keyup/change events.
Advanced: Add custom CAPTCHA using Canvas.

code:

Style.css

```
/* =====  
Practical 5 – Registration Form with  
Frontend Validation & Error Handling  
===== */  
  
:root {  
  --primary: #1a73e8;  
  --secondary: #0b3d91;  
  --accent: #fbbc04;  
  --success: #16a34a;  
  --danger: #dc2626;  
  --warning: #d97706;  
  --dark: #1e2a3a;  
  --light: #ffffff;
```

```
--bg:      #f0f4f9;
--surface: #ffffff;
--muted:   #6b7280;
--border:  #d1d5db;
--border-ok: #16a34a;
--border-err: #dc2626;
--radius:  8px;
--shadow:   0 4px 24px rgba(0,0,0,.12);
--trans:    0.22s ease;
--font:     'Segoe UI', system-ui, sans-serif;
}
```

```
*, *::before, *::after { box-sizing: border-box; margin: 0; padding: 0; }
html { scroll-behavior: smooth; font-size: 16px; }
body {
  font-family: var(--font);
  background: var(--bg);
  color: var(--dark);
  line-height: 1.65;
  min-height: 100vh;
}
```

```
/* ---- Skip Link ---- */
```

```
.skip-link {
  position: absolute; top: -50px; left: 0;
  background: var(--primary); color: #fff;
  padding: .5rem 1rem; z-index: 9999;
  text-decoration: none; font-weight: 600;
  border-radius: 0 0 4px 0;
  transition: top var(--trans);
}
.skip-link:focus { top: 0; }
```

```
/* ---- Navbar ---- */
```

```
.navbar {
  background: var(--secondary);
  padding: 0 1.5rem;
  display: flex; align-items: center; justify-content: space-between;
  height: 62px; position: sticky; top: 0; z-index: 100;
  box-shadow: 0 2px 8px rgba(0,0,0,.2);
}
.navbar .logo { font-size: 1.35rem; font-weight: 700; color: var(--accent); text-decoration: none; }
.navbar nav ul { display: flex; gap: 1.2rem; list-style: none; }
```

```
.navbar nav a { color: #c8d9f5; text-decoration: none; font-size: .9rem; }  
.navbar nav a:hover { color: var(--accent); }
```

```
/* ---- Page wrapper ---- */
```

```
.page-wrap {  
  display: flex; align-items: flex-start; justify-content: center;  
  gap: 2rem; padding: 2.5rem 1.5rem;  
  max-width: 1100px; margin: 0 auto;  
}
```

```
/* ---- Form Card ---- */
```

```
.form-card {  
  background: var(--surface);  
  border-radius: 12px;  
  padding: 2.2rem;  
  box-shadow: var(--shadow);  
  flex: 1 1 0;  
  max-width: 680px;  
  width: 100%;  
}  
.form-card h1 {  
  font-size: 1.6rem; color: var(--secondary);  
  margin-bottom: .3rem;  
}  
.form-card .subtitle { font-size: .9rem; color: var(--muted); margin-bottom: 1.8rem; }
```

```
/* ---- Progress Steps ---- */
```

```
.steps {  
  display: flex; gap: 0; margin-bottom: 2rem;  
  counter-reset: step;  
}  
.step {  
  flex: 1; text-align: center; position: relative; font-size: .78rem;  
  color: var(--muted);  
}  
.step::before {  
  counter-increment: step;  
  content: counter(step);  
  display: block; width: 28px; height: 28px;  
  border-radius: 50%; border: 2px solid var(--border);  
  background: var(--surface); line-height: 24px;  
  margin: 0 auto .4rem; font-weight: 700; font-size: .82rem;  
  transition: background var(--trans), border-color var(--trans), color var(--trans);  
  color: var(--muted);
```

```

}
.step::after {
  content: ""; position: absolute;
  top: 14px; left: 50%; right: -50%;
  height: 2px; background: var(--border);
  z-index: -1;
}
.step:last-child::after { display: none; }
.step.active::before { background: var(--primary); border-color: var(--primary); color:
#fff; }
.step.completed::before { background: var(--success); border-color: var(--success);
color: #fff; content: '✓'; }
.step.active { color: var(--primary); font-weight: 600; }
.step.completed { color: var(--success); }

/* ---- Fieldsets ---- */
fieldset { border: none; margin-bottom: 1.4rem; }
legend {
  font-size: 1rem; font-weight: 700;
  color: var(--secondary); margin-bottom: 1rem;
  padding-bottom: .4rem;
  border-bottom: 2px solid var(--accent);
  width: 100%; display: block;
}

/* ---- Field Groups ---- */
.field-row {
  display: grid; grid-template-columns: 1fr 1fr; gap: 1rem;
}
@media (max-width: 560px) { .field-row { grid-template-columns: 1fr; } }

.field-group { margin-bottom: 1.1rem; position: relative; }

.field-group label {
  display: block; font-weight: 600; font-size: .9rem;
  margin-bottom: .35rem; color: var(--secondary);
}
.field-group label .required-star {
  color: var(--danger); margin-left: 2px; font-size: .85rem;
}

/* Input base */
.field-group input,
.field-group select,

```

```

.field-group textarea {
  width: 100%;
  padding: .6rem .9rem;
  border: 1.5px solid var(--border);
  border-radius: var(--radius);
  font-size: .97rem; font-family: var(--font);
  background: #fafafa;
  transition: border-color var(--trans), box-shadow var(--trans), background var(--trans);
  color: var(--dark);
}
.field-group input:focus,
.field-group select:focus,
.field-group textarea:focus {
  outline: none;
  border-color: var(--primary);
  box-shadow: 0 0 0 3px rgba(26,115,232,.18);
  background: #fff;
}

/* Valid state */
.field-group.valid input,
.field-group.valid select,
.field-group.valid textarea {
  border-color: var(--border-ok);
  background: #f0fdf4;
}

/* Invalid state */
.field-group.invalid input,
.field-group.invalid select,
.field-group.invalid textarea {
  border-color: var(--border-err);
  background: #fff5f5;
}

/* Validation icon inside input */
.field-group .val-icon {
  position: absolute; right: .8rem; top: 2.3rem;
  font-size: 1rem; pointer-events: none;
  opacity: 0; transition: opacity var(--trans);
}
.field-group.valid .val-icon { opacity: 1; }
.field-group.invalid .val-icon { opacity: 1; }

/* Error message */

```

```

.err-msg {
  font-size: .8rem; color: var(--danger);
  margin-top: .3rem; display: none;
  align-items: center; gap: .3rem;
}
.err-msg.show { display: flex; }
.err-msg::before { content: '△'; font-size: .85rem; }

/* Hint text */
.hint {
  font-size: .78rem; color: var(--muted);
  margin-top: .25rem; display: block;
}

/* ---- Password Wrapper (show/hide toggle) ---- */
.pwd-wrap { position: relative; }
.pwd-wrap input { padding-right: 2.8rem; }
.pwd-toggle {
  position: absolute; right: .7rem; top: 50%; transform: translateY(-50%);
  background: none; border: none; cursor: pointer;
  font-size: 1rem; color: var(--muted);
  transition: color var(--trans); padding: 0;
}
.pwd-toggle:hover { color: var(--primary); }

/* ---- Password Strength Meter ---- */
.strength-bar-wrap {
  display: grid; grid-template-columns: repeat(4, 1fr); gap: 4px;
  margin-top: .5rem;
}
.strength-segment {
  height: 5px; border-radius: 999px;
  background: var(--border);
  transition: background var(--trans);
}
.strength-label {
  font-size: .78rem; font-weight: 600;
  margin-top: .3rem; color: var(--muted);
  transition: color var(--trans);
}

/* Strength levels */
[data-strength="1"] .strength-segment:nth-child(1) { background: var(--danger); }
[data-strength="2"] .strength-segment:nth-child(-n+2) { background: var(--warning); }

```

```
[data-strength="3"] .strength-segment:nth-child(-n+3) { background: var(--accent); }  
[data-strength="4"] .strength-segment { background: var(--success); }
```

```
[data-strength="1"] .strength-label { color: var(--danger); }  
[data-strength="2"] .strength-label { color: var(--warning); }  
[data-strength="3"] .strength-label { color: #b45309; }  
[data-strength="4"] .strength-label { color: var(--success); }
```

```
/* ---- Captcha (Canvas) ---- */
```

```
.captcha-wrap {  
  display: flex; align-items: center; gap: .8rem; flex-wrap: wrap;  
  margin-bottom: .5rem;  
}  
#captcha-canvas {  
  border: 1.5px solid var(--border); border-radius: var(--radius);  
  cursor: pointer; background: #fff;  
}  
#captcha-canvas:hover { opacity: .85; }  
#captcha-refresh {  
  background: none; border: none; cursor: pointer;  
  font-size: 1.1rem; color: var(--primary);  
  transition: transform .4s ease;  
}  
#captcha-refresh:hover { transform: rotate(180deg); }
```

```
/* ---- Checkbox / Radio ---- */
```

```
.check-label {  
  display: flex; gap: .5rem; align-items: flex-start;  
  cursor: pointer; font-size: .9rem;  
}  
.check-label input[type="checkbox"],  
.check-label input[type="radio"] { width: auto; margin-top: .25rem; flex-shrink: 0; }
```

```
.gender-group { display: flex; gap: 1.2rem; flex-wrap: wrap; }
```

```
/* ---- Submit Button ---- */
```

```
.submit-btn {  
  display: block; width: 100%;  
  padding: .8rem; font-size: 1.05rem; font-weight: 700;  
  background: linear-gradient(135deg, var(--primary), var(--secondary));  
  color: #fff; border: none; border-radius: var(--radius);  
  cursor: pointer; letter-spacing: .5px;  
  transition: opacity var(--trans), transform var(--trans);  
  margin-top: .5rem;
```

```

}
.submit-btn:hover { opacity: .92; transform: translateY(-2px); }
.submit-btn:active { transform: translateY(0); }
.submit-btn:disabled { opacity: .5; cursor: not-allowed; transform: none; }

/* ---- Success Panel ---- */
#success-panel {
  display: none; text-align: center; padding: 2rem 1rem;
}
#success-panel .success-icon { font-size: 4rem; margin-bottom: 1rem; }
#success-panel h2 { color: var(--success); margin-bottom: .5rem; }
#success-panel p { color: var(--muted); font-size: .95rem; }

/* ---- Summary Sidebar ---- */
.summary-card {
  background: var(--surface); border-radius: 12px;
  padding: 1.5rem; box-shadow: var(--shadow);
  width: 280px; flex-shrink: 0;
  position: sticky; top: 80px;
  align-self: flex-start;
}
.summary-card h2 { font-size: 1rem; font-weight: 700; color: var(--secondary);
margin-bottom: 1rem; }
.summary-row {
  display: flex; justify-content: space-between;
  font-size: .85rem; padding: .4rem 0;
  border-bottom: 1px solid var(--border);
  gap: .5rem;
}
.summary-row:last-child { border-bottom: none; }
.summary-row .s-label { color: var(--muted); flex-shrink: 0; }
.summary-row .s-val { font-weight: 600; text-align: right; word-break: break-all; }
.requirements-list { list-style: none; font-size: .8rem; margin-top: .8rem; }
.requirements-list li { padding: .25rem 0; display: flex; align-items: center; gap: .4rem; }
.requirements-list .req-icon { font-size: .9rem; }
.req-met { color: var(--success); }
.req-unmet { color: var(--muted); }

@media (max-width: 900px) {
  .page-wrap { flex-direction: column; align-items: center; }
  .summary-card { width: 100%; max-width: 680px; position: static; }
}

/* ---- Footer ---- */

```

```

.site-footer {
  background: var(--secondary); color: #94a3b8;
  text-align: center; padding: 1.2rem;
  font-size: .85rem; margin-top: 3rem;
}
.site-footer a { color: var(--accent); text-decoration: none; }

/* ---- Utility ---- */
.visually-hidden {
  position: absolute; width: 1px; height: 1px; padding: 0;
  margin: -1px; overflow: hidden; clip: rect(0,0,0,0); border: 0;
}

```

validate.js

```

/**
 * Practical 5 – Registration Form Validation
 * Features:
 * - Regex validation for all fields
 * - Real-time keyup/change validation (Intermediate)
 * - Password strength meter
 * - Show/hide password toggle
 * - Custom Canvas CAPTCHA (Advanced)
 * - Accessible error messages (aria-describedby)
 * - Live summary sidebar update
 * - Step progress indicator
 * - Form submission prevention until valid
 */

'use strict';

/* =====
  REGEX PATTERNS
  ===== */
const PATTERNS = {
  firstName: /^[A-Za-z][A-Za-z\s']{1,49}$/,
  lastName: /^[A-Za-z][A-Za-z\s']{1,49}$/,
  email: /^[a-zA-Z0-9._%+\-]+@[a-zA-Z0-9.\-]+\.[a-zA-Z]{2,}$/,
  mobile: /^[6-9]\d{9}$/, // Indian 10-digit
  dob: /^[d{4}-\d{2}-\d{2}]$/,
  password: /^(?=.*[a-z])(?=.*[A-Z])(?=.*\d)(?=.*[@$!%*?&_#^])[A-Za-z\d@$!%*?&_#^]{8,}$/,
  rollNumber: /^[A-Z]{2,5}\d{4}\d{3,6}$/i, // e.g. BCA/2024/021
};

const ERROR_MSGS = {

```

```

firstName: 'First name must be 2–50 letters (hyphens and apostrophes allowed).',
lastName: 'Last name must be 2–50 letters (hyphens and apostrophes allowed).',
email: 'Enter a valid email address (e.g. student@example.com).',
mobile: 'Enter a valid 10-digit Indian mobile number starting with 6–9.',
dob: 'Date of birth is required. You must be at least 15 years old.',
course: 'Please select a course.',
year: 'Please select your year of study.',
gender: 'Please select a gender.',
password: 'Min 8 chars with uppercase, lowercase, number, and special character
(@$!%*?&_#^).',
confirmPwd: 'Passwords do not match.',
rollNumber: 'Format: BCA/2024/021 (Course/Year/Number).',
terms: 'You must accept the Terms & Conditions to register.',
captcha: 'Incorrect CAPTCHA. Click the image to refresh and try again.',
};

```

```

/* =====
UTILITY HELPERS
===== */
function $(id) { return document.getElementById(id); }

```

```

/**
 * Mark a field as valid or invalid.
 * @param {HTMLElement} group - .field-group element
 * @param {boolean}   isOk
 * @param {string}    [msg] - Error message when invalid
 */
function setValidity(group, isOk, msg = "") {
  const errEl = group.querySelector('.err-msg');
  const icon = group.querySelector('.val-icon');

  group.classList.toggle('valid', isOk);
  group.classList.toggle('invalid', !isOk);

  const input = group.querySelector('input, select, textarea');
  if (input) {
    input.setAttribute('aria-invalid', String(!isOk));
  }

  if (errEl) {
    errEl.textContent = isOk ? "" : msg;
    errEl.classList.toggle('show', !isOk);
  }
  if (icon) {
    icon.textContent = isOk ? '✓' : '✗';
  }
}

/** Clear validation state (neutral) */

```

```

function clearValidity(group) {
  group.classList.remove('valid', 'invalid');
  const errEl = group.querySelector('.err-msg');
  const icon = group.querySelector('.val-icon');
  if (errEl) { errEl.textContent = ""; errEl.classList.remove('show'); }
  if (icon) { icon.textContent = ""; }
  const input = group.querySelector('input, select, textarea');
  if (input) input.removeAttribute('aria-invalid');
}

/** Get .field-group ancestor of an element */
function getGroup(el) { return el.closest('.field-group'); }

/* =====
  INDIVIDUAL FIELD VALIDATORS
  ===== */
function validateFirstName() {
  const el = $('first-name');
  const group = getGroup(el);
  const v = el.value.trim();
  const ok = PATTERNS.firstName.test(v);
  setValidity(group, ok, ERROR_MSGS.firstName);
  updateSummary('s-fname', v || '—');
  return ok;
}

function validateLastName() {
  const el = $('last-name');
  const group = getGroup(el);
  const v = el.value.trim();
  const ok = PATTERNS.lastName.test(v);
  setValidity(group, ok, ERROR_MSGS.lastName);
  updateSummary('s-lname', v || '—');
  return ok;
}

function validateEmail() {
  const el = $('email');
  const group = getGroup(el);
  const v = el.value.trim();
  const ok = PATTERNS.email.test(v);
  setValidity(group, ok, ERROR_MSGS.email);
  updateSummary('s-email', v || '—');
  return ok;
}

function validateMobile() {
  const el = $('mobile');
  const group = getGroup(el);

```

```

const v    = el.value.trim();
const ok   = PATTERNS.mobile.test(v);
setValidity(group, ok, ERROR_MSGS.mobile);
return ok;
}

```

```

function validateDOB() {
  const el   = $('dob');
  const group = getGroup(el);
  const v    = el.value;
  if (!v) { setValidity(group, false, ERROR_MSGS.dob); return false; }
  const dob  = new Date(v);
  const today = new Date();
  const age  = (today - dob) / (365.25 * 24 * 3600 * 1000);
  const ok   = !isNaN(dob) && age >= 15;
  setValidity(group, ok, 'You must be at least 15 years old.');
```

```

  return ok;
}

```

```

function validateCourse() {
  const el   = $('course');
  const group = getGroup(el);
  const ok   = el.value !== "";
  setValidity(group, ok, ERROR_MSGS.course);
  updateSummary('s-course', ok ? el.options[el.selectedIndex].text : '—');
  return ok;
}

```

```

function validateYear() {
  const el   = $('year');
  const group = getGroup(el);
  const ok   = el.value !== "";
  setValidity(group, ok, ERROR_MSGS.year);
  return ok;
}

```

```

function validateGender() {
  const selected = document.querySelector('input[name="gender"]:checked');
  const group    = document.getElementById('gender-group-wrap');
  const ok       = !!selected;
  if (group) {
    group.classList.toggle('invalid', !ok);
    const errEl = group.querySelector('.err-msg');
    if (errEl) { errEl.textContent = ok ? "" : ERROR_MSGS.gender; errEl.classList.toggle('show', !ok); }
  }
  return ok;
}

```

```
function validateRollNumber() {
  const el = $('roll-number');
  const group = getGroup(el);
  const v = el.value.trim();
  const ok = v === "" || PATTERNS.rollNumber.test(v); // optional field
  setValidity(group, ok, ERROR_MSGS.rollNumber);
  return ok;
}
```

```
/* ---- Password Strength ---- */
```

```
/**
```

```
 * Calculate password strength score (0–4).
```

```
 * @param {string} pwd
```

```
 * @returns {{ score: number, label: string }}
```

```
 */
```

```
function getPasswordStrength(pwd) {
```

```
  let score = 0;
```

```
  if (pwd.length >= 8)      score++;
```

```
  if (/[A-Z]/.test(pwd))    score++;
```

```
  if (/^d/.test(pwd))      score++;
```

```
  if (/[@$!%*?&_#^]/.test(pwd)) score++;
```

```
  const labels = ['', 'Weak', 'Fair', 'Good', 'Strong'];
```

```
  return { score, label: labels[score] || '' };
```

```
}
```

```
function updateStrengthMeter(pwd) {
```

```
  const wrap = document.getElementById('strength-wrap');
```

```
  if (!wrap) return;
```

```
  const { score, label } = getPasswordStrength(pwd);
```

```
  wrap.dataset.strength = score;
```

```
  const labelEl = document.getElementById('strength-label');
```

```
  if (labelEl) labelEl.textContent = pwd ? `Strength: ${label}` : '';
```

```
  // Update password requirements checklist
```

```
  updateReqItem('req-len',    pwd.length >= 8);
```

```
  updateReqItem('req-upper',  /[A-Z]/.test(pwd));
```

```
  updateReqItem('req-lower',  /[a-z]/.test(pwd));
```

```
  updateReqItem('req-num',    /^d/.test(pwd));
```

```
  updateReqItem('req-special', /[@$!%*?&_#^]/.test(pwd));
```

```
}
```

```
function updateReqItem(id, met) {
```

```
  const el = document.getElementById(id);
```

```
  if (!el) return;
```

```
  const icon = el.querySelector('.req-icon');
```

```
  el.classList.toggle('req-met', met);
```

```
  el.classList.toggle('req-unmet', !met);
```

```
  if (icon) icon.textContent = met ? '✔' : '○';
```

```
}
```

```
function validatePassword() {  
  const el = $('password');  
  const group = getGroup(el);  
  const v = el.value;  
  const ok = PATTERNS.password.test(v);  
  setValidity(group, ok, ERROR_MSGS.password);  
  updateStrengthMeter(v);  
  // Re-validate confirm if already touched  
  if ($('confirm-pwd').value) validateConfirmPwd();  
  return ok;  
}
```

```
function validateConfirmPwd() {  
  const el = $('confirm-pwd');  
  const group = getGroup(el);  
  const ok = el.value === $('password').value && el.value !== "";  
  setValidity(group, ok, ERROR_MSGS.confirmPwd);  
  return ok;  
}
```

```
function validateTerms() {  
  const el = $('terms');  
  const group = getGroup(el);  
  const ok = el.checked;  
  group.classList.toggle('invalid', !ok);  
  const errEl = group.querySelector('.err-msg');  
  if (errEl) { errEl.textContent = ok ? "" : ERROR_MSGS.terms; errEl.classList.toggle('show', !ok); }  
  return ok;  
}
```

```
/* =====  
  CAPTCHA (Advanced Extension – Canvas-based)  
  ===== */
```

```
let captchaCode = "";
```

```
function generateCaptcha() {  
  const canvas = $('captcha-canvas');  
  if (!canvas) return;  
  const ctx = canvas.getContext('2d');  
  const chars = 'ABCDEFGHJKLMNPQRSTUVWXYZ23456789';  
  captchaCode = Array.from({ length: 6 }, () => chars[Math.floor(Math.random() *  
  chars.length)]).join("");
```

```
  const W = canvas.width, H = canvas.height;  
  ctx.clearRect(0, 0, W, H);
```

```
  // Background gradient
```

```

const grad = ctx.createLinearGradient(0, 0, W, H);
grad.addColorStop(0, '#e0e7ff'); grad.addColorStop(1, '#f0f4f9');
ctx.fillStyle = grad;
ctx.fillRect(0, 0, W, H);

// Noise lines
for (let i = 0; i < 6; i++) {
  ctx.strokeStyle = `hsla(${Math.random()*360},60%,65%,.5)`;
  ctx.lineWidth = 1;
  ctx.beginPath();
  ctx.moveTo(Math.random() * W, Math.random() * H);
  ctx.lineTo(Math.random() * W, Math.random() * H);
  ctx.stroke();
}

// Noise dots
for (let i = 0; i < 40; i++) {
  ctx.fillStyle = `hsla(${Math.random()*360},60%,65%,.4)`;
  ctx.beginPath();
  ctx.arc(Math.random() * W, Math.random() * H, 1.5, 0, Math.PI * 2);
  ctx.fill();
}

// Draw characters with slight rotation
const fonts = ['bold 22px Georgia', 'bold 22px Arial', 'bold 22px Courier New'];
const colors = ['#1a237e', '#b71c1c', '#1b5e20', '#4a148c', '#e65100', '#006064'];
captchaCode.split('').forEach((ch, i) => {
  ctx.save();
  ctx.font = fonts[i % fonts.length];
  ctx.fillStyle = colors[i % colors.length];
  const x = 14 + i * 26;
  const y = H / 2 + 8;
  ctx.translate(x, y);
  ctx.rotate((Math.random() - 0.5) * 0.5);
  ctx.fillText(ch, 0, 0);
  ctx.restore();
});

// Update aria-label for screen readers (provide alternative accessible code)
canvas.setAttribute('aria-label', `CAPTCHA image. The code is: ${captchaCode.split('').join(' ')}`);
}

function validateCaptcha() {
  const el = $('captcha-input');
  const group = getGroup(el);
  const ok = el.value.trim().toUpperCase() === captchaCode;
  setValidity(group, ok, ERROR_MSGS.captcha);
  return ok;
}

```

```

/* =====
SHOW / HIDE PASSWORD TOGGLE
===== */
function setupPasswordToggle(inputId, btnId) {
  const input = $(inputId);
  const btn = $(btnId);
  if (!input || !btn) return;
  btn.addEventListener('click', () => {
    const isText = input.type === 'text';
    input.type = isText ? 'password' : 'text';
    btn.textContent = isText ? '👁️' : '🔒';
    btn.setAttribute('aria-label', isText ? 'Show password' : 'Hide password');
  });
}

/* =====
SUMMARY SIDEBAR UPDATE
===== */
function updateSummary(id, value) {
  const el = $(id);
  if (el) el.textContent = value;
}

/* =====
STEP PROGRESS INDICATOR
===== */
const fieldStepMap = {
  'first-name': 1, 'last-name': 1, 'email': 1, 'mobile': 1, 'dob': 1,
  'course': 2, 'year': 2, 'roll-number': 2,
  'password': 3, 'confirm-pwd': 3,
};

function updateSteps() {
  const steps = document.querySelectorAll('.step');
  const allOk = [
    validateFirstName, validateLastName, validateEmail, validateMobile, validateDOB,
    validateCourse, validateYear, validateGender,
    validatePassword, validateConfirmPwd,
    validateTerms, validateCaptcha
  ];
  // Just use visual indication based on filled fields (non-destructive)
  const step1Fields = ['first-name', 'last-name', 'email', 'mobile', 'dob'];
  const step2Fields = ['course', 'year'];
  const step3Fields = ['password', 'confirm-pwd'];

  function allFilled(ids) { return ids.every(id => { const el = $(id); return el && el.value.trim(); }); }

  if (steps[0]) steps[0].classList.toggle('completed', allFilled(step1Fields));

```

```

    if (steps[1]) steps[1].classList.toggle('completed', allFilled(step1Fields) && allFilled(step2Fields));
    if (steps[2]) steps[2].classList.toggle('completed', allFilled(step1Fields) && allFilled(step2Fields)
&& allFilled(step3Fields) && $('terms')?.checked);

```

```

// Active step
if (!allFilled(step1Fields)) {
    steps.forEach((s, i) => s.classList.toggle('active', i === 0));
} else if (!allFilled(step2Fields)) {
    steps.forEach((s, i) => s.classList.toggle('active', i === 1));
} else {
    steps.forEach((s, i) => s.classList.toggle('active', i === 2));
}
}

```

```

/* =====
REAL-TIME VALIDATION (Intermediate Extension)
===== */

```

```

function attachRealTimeValidation() {

```

```

    const map = [
        { id: 'first-name', fn: validateFirstName, events: ['input', 'blur'] },
        { id: 'last-name', fn: validateLastName, events: ['input', 'blur'] },
        { id: 'email', fn: validateEmail, events: ['input', 'blur'] },
        { id: 'mobile', fn: validateMobile, events: ['input', 'blur'] },
        { id: 'dob', fn: validateDOB, events: ['change', 'blur'] },
        { id: 'course', fn: validateCourse, events: ['change'] },
        { id: 'year', fn: validateYear, events: ['change'] },
        { id: 'roll-number', fn: validateRollNumber, events: ['input', 'blur'] },
        { id: 'password', fn: validatePassword, events: ['input', 'blur'] },
        { id: 'confirm-pwd', fn: validateConfirmPwd, events: ['input', 'blur'] },
        { id: 'captcha-input', fn: validateCaptcha, events: ['input', 'blur'] },
    ];

```

```

    map.forEach(({ id, fn, events }) => {
        const el = $(id);
        if (!el) return;
        events.forEach(ev => {
            el.addEventListener(ev, () => { fn(); updateSteps(); });
        });
    });

```

```

// Gender radios
document.querySelectorAll('input[name="gender"]').forEach(r => {
    r.addEventListener('change', () => { validateGender(); updateSteps(); });
});

```

```

// Terms
const termsEl = $('terms');
if (termsEl) termsEl.addEventListener('change', () => { validateTerms(); updateSteps(); });

```

```
}
```

```
/* =====  
FULL FORM VALIDATION (on submit)  
===== */
```

```
function validateAll() {  
  const results = [  
    validateFirstName(),  
    validateLastName(),  
    validateEmail(),  
    validateMobile(),  
    validateDOB(),  
    validateCourse(),  
    validateYear(),  
    validateGender(),  
    validateRollNumber(),  
    validatePassword(),  
    validateConfirmPwd(),  
    validateTerms(),  
    validateCaptcha(),  
  ];  
  return results.every(Boolean);  
}
```

```
/* =====  
FORM SUBMIT HANDLER  
===== */
```

```
function handleSubmit(e) {  
  e.preventDefault();  
  if (validateAll()) {  
    // Show success panel  
    document.getElementById('form-section').style.display = 'none';  
    const panel = document.getElementById('success-panel');  
    if (panel) {  
      panel.style.display = 'block';  
      panel.focus();  
    }  
  } else {  
    // Scroll to first invalid field  
    const firstInvalid = document.querySelector('.field-group.invalid input, .field-group.invalid select');  
    if (firstInvalid) {  
      firstInvalid.scrollIntoView({ behavior: 'smooth', block: 'center' });  
      firstInvalid.focus();  
    }  
  }  
}
```

```
/* =====
```

```

INIT
===== */
document.addEventListener('DOMContentLoaded', () => {

  // Captcha
  generateCaptcha();
  const refreshBtn = $('captcha-refresh');
  const canvas = $('captcha-canvas');
  if (refreshBtn) refreshBtn.addEventListener('click', () => { generateCaptcha(); if
$('captcha-input') $('captcha-input').value = ""; clearValidity(getGroup($('captcha-input'))); });
  if (canvas) canvas.addEventListener('click', () => { generateCaptcha(); if $('captcha-input')
$('captcha-input').value = ""; clearValidity(getGroup($('captcha-input'))); });

  // Show/hide password toggles
  setupPasswordToggle('password', 'pwd-toggle-1');
  setupPasswordToggle('confirm-pwd', 'pwd-toggle-2');

  // Real-time validation
  attachRealTimeValidation();

  // Form submit
  const form = $('registration-form');
  if (form) form.addEventListener('submit', handleSubmit);

  // Reset button
  const resetBtn = $('reset-btn');
  if (resetBtn) {
    resetBtn.addEventListener('click', () => {
      document.querySelectorAll('.field-group').forEach(g => clearValidity(g));
      updateStrengthMeter("");
      generateCaptcha();
      updateSteps();
      // Clear summary
      ['s-fname', 's-lname', 's-email', 's-course'].forEach(id => updateSummary(id, '—'));
    });
  }
});

```

Index.html

```

<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8" />
  <meta name="viewport" content="width=device-width, initial-scale=1.0" />
  <meta name="description" content="Practical 5 – Student Registration Form with
JavaScript Validation, Regex, and Password Strength Meter." />

```

```

<title>Practical 5 – Registration Form | StudentHub</title>
<link rel="stylesheet" href="css/style.css" />
</head>
<body>

  <a class="skip-link" href="#main-content">Skip to main content</a>

  <!-- NAVBAR -->
  <header>
    <div class="navbar" role="banner">
      <a href="../../practical2/index.html" class="logo" aria-label="StudentHub Home">🎓
StudentHub</a>
      <nav aria-label="Primary navigation">
        <ul>
          <li><a href="../../practical2/index.html">Home</a></li>
          <li><a href="../../practical4/index.html">Practical 4</a></li>
          <li><a href="../../practical2/login.html">Login</a></li>
        </ul>
      </nav>
    </div>
  </header>

  <main id="main-content">
    <div class="page-wrap">

      <!-- =====
      REGISTRATION FORM CARD
      ===== -->
      <div class="form-card">

        <h1>Create Your Account</h1>
        <p class="subtitle">
          Practical 5 – HTML5 inputs · JS Regex Validation · Real-time Feedback ·
          Password Strength Meter · Canvas CAPTCHA
        </p>
        <p class="subtitle" style="margin-top: -.8rem;">
          Already registered? <a href="../../practical2/login.html">Login here</a>.
        </p>

        <!-- Step Progress Indicator -->
        <ol class="steps" aria-label="Registration progress">
          <li class="step active">Personal Info</li>
          <li class="step">Academic</li>
          <li class="step">Security</li>

```


<!-- ===== FORM ===== -->

<form id="registration-form" action="#" method="post" novalidate
aria-label="Student registration form" id="form-section">

<!-- ---- SECTION 1: Personal Information ---- -->

<fieldset>

<legend>Personal Information</legend>

<div class="field-row">

<!-- First Name -->

<div class="field-group" id="fg-fname">

<label for="first-name">

First Name *

</label>

<input type="text" id="first-name" name="firstName"

autocomplete="given-name"

required aria-required="true"

aria-describedby="err-fname"

placeholder="e.g. Rahul" />

</div>

<!-- Last Name -->

<div class="field-group" id="fg-lname">

<label for="last-name">

Last Name *

</label>

<input type="text" id="last-name" name="lastName"

autocomplete="family-name"

required aria-required="true"

aria-describedby="err-lname"

placeholder="e.g. Verma" />

</div>

</div>

<!-- Email -->

<div class="field-group" id="fg-email">

<label for="email">

Email Address *

```

</label>
<input type="email" id="email" name="email"
  autocomplete="email"
  required aria-required="true"
  aria-describedby="err-email"
  placeholder="student@example.com" />
<span class="val-icon" aria-hidden="true"></span>
<span class="err-msg" id="err-email" role="alert"></span>
</div>

<div class="field-row">
  <!-- Mobile -->
  <div class="field-group" id="fg-mobile">
    <label for="mobile">
      Mobile Number <span class="required-star" aria-hidden="true">*</span>
    </label>
    <input type="tel" id="mobile" name="mobile"
      autocomplete="tel"
      required aria-required="true"
      aria-describedby="err-mobile"
      placeholder="10-digit number"
      maxlength="10" inputmode="numeric" />
    <span class="val-icon" aria-hidden="true"></span>
    <span class="err-msg" id="err-mobile" role="alert"></span>
    <span class="hint">Indian format: starts with 6–9</span>
  </div>

  <!-- Date of Birth -->
  <div class="field-group" id="fg-dob">
    <label for="dob">
      Date of Birth <span class="required-star" aria-hidden="true">*</span>
    </label>
    <input type="date" id="dob" name="dob"
      required aria-required="true"
      aria-describedby="err-dob"
      max="2011-12-31" />
    <span class="val-icon" aria-hidden="true"></span>
    <span class="err-msg" id="err-dob" role="alert"></span>
  </div>
</div>

  <!-- Gender (radio group) -->
  <div class="field-group" id="gender-group-wrap">
    <fieldset>

```

```

        <legend>Gender <span class="required-star"
aria-hidden="true">*</span></legend>
        <div class="gender-group" role="radiogroup" aria-required="true">
            <label class="check-label">
                <input type="radio" name="gender" value="male" required /> Male
            </label>
            <label class="check-label">
                <input type="radio" name="gender" value="female" /> Female
            </label>
            <label class="check-label">
                <input type="radio" name="gender" value="other" /> Other / Prefer not to say
            </label>
        </div>
    </fieldset>
    <span class="err-msg" id="err-gender" role="alert"></span>
</div>

```

```

</fieldset>

```

```

<!-- ---- SECTION 2: Academic Information ---- -->

```

```

<fieldset>

```

```

    <legend>Academic Information</legend>

```

```

    <div class="field-row">

```

```

        <!-- Course -->

```

```

        <div class="field-group" id="fg-course">

```

```

            <label for="course">

```

```

                Course <span class="required-star" aria-hidden="true">*</span>

```

```

            </label>

```

```

            <select id="course" name="course"

```

```

                required aria-required="true"

```

```

                aria-describedby="err-course">

```

```

                <option value="">-- Select Course --</option>

```

```

                <option value="bsc-cs">B.Sc. Computer Science</option>

```

```

                <option value="bca">BCA</option>

```

```

                <option value="mca">MCA</option>

```

```

                <option value="msc-cs">M.Sc. Computer Science</option>

```

```

                <option value="bcom">B.Com</option>

```

```

                <option value="bba">BBA</option>

```

```

            </select>

```

```

            <span class="val-icon" aria-hidden="true"></span>

```

```

            <span class="err-msg" id="err-course" role="alert"></span>

```

```

        </div>

```

```

<!-- Year -->
<div class="field-group" id="fg-year">
  <label for="year">
    Year of Study <span class="required-star" aria-hidden="true">*</span>
  </label>
  <select id="year" name="year"
    required aria-required="true"
    aria-describedby="err-year">
    <option value="">-- Select Year --</option>
    <option value="1">First Year</option>
    <option value="2">Second Year</option>
    <option value="3">Third Year</option>
    <option value="4">Fourth Year (PG)</option>
  </select>
  <span class="val-icon" aria-hidden="true"></span>
  <span class="err-msg" id="err-year" role="alert"></span>
</div>
</div>

<!-- Roll Number (optional) -->
<div class="field-group" id="fg-roll">
  <label for="roll-number">Roll Number <small>(optional)</small></label>
  <input type="text" id="roll-number" name="rollNumber"
    aria-describedby="err-roll hint-roll"
    placeholder="e.g. BCA/2024/021" />
  <span class="val-icon" aria-hidden="true"></span>
  <span class="err-msg" id="err-roll" role="alert"></span>
  <span class="hint" id="hint-roll">Format: COURSE/YEAR/NUMBER (e.g.
BCA/2024/021)</span>
</div>

</fieldset>

<!-- ---- SECTION 3: Account Security ---- -->
<fieldset>
  <legend>Account Security</legend>

  <!-- Password -->
  <div class="field-group" id="fg-pwd">
    <label for="password">
      Password <span class="required-star" aria-hidden="true">*</span>
    </label>
    <div class="pwd-wrap">
      <input type="password" id="password" name="password"

```

```

        autocomplete="new-password"
        required aria-required="true"
        aria-describedby="err-pwd strength-label"
        placeholder="Create a strong password" />
        <button type="button" class="pwd-toggle" id="pwd-toggle-1"
            aria-label="Show password">👁️</button>
    </div>
    <span class="val-icon" aria-hidden="true"></span>
    <span class="err-msg" id="err-pwd" role="alert"></span>

    <!-- Strength Meter -->
    <div id="strength-wrap" data-strength="0" aria-hidden="true">
        <div class="strength-bar-wrap">
            <div class="strength-segment"></div>
            <div class="strength-segment"></div>
            <div class="strength-segment"></div>
            <div class="strength-segment"></div>
        </div>
        <span class="strength-label" id="strength-label"></span>
    </div>

    <!-- Password Requirements Checklist -->
    <ul class="requirements-list" aria-label="Password requirements">
        <li id="req-len" class="req-unmet"><span class="req-icon">○</span> At least
8 characters</li>
        <li id="req-upper" class="req-unmet"><span class="req-icon">○</span> At
least one uppercase letter</li>
        <li id="req-lower" class="req-unmet"><span class="req-icon">○</span> At
least one lowercase letter</li>
        <li id="req-num" class="req-unmet"><span class="req-icon">○</span> At
least one number</li>
        <li id="req-special" class="req-unmet"><span class="req-icon">○</span> At
least one special character (@$!%*?&amp;_#^)</li>
    </ul>
</div>

    <!-- Confirm Password -->
    <div class="field-group" id="fg-cpwd">
        <label for="confirm-pwd">
            Confirm Password <span class="required-star" aria-hidden="true">*</span>
        </label>
        <div class="pwd-wrap">
            <input type="password" id="confirm-pwd" name="confirmPassword"
                autocomplete="new-password"

```

```

        required aria-required="true"
        aria-describedby="err-cpwd"
        placeholder="Re-enter your password" />
    <button type="button" class="pwd-toggle" id="pwd-toggle-2"
        aria-label="Show password">👁️</button>
</div>
<span class="val-icon" aria-hidden="true"></span>
<span class="err-msg" id="err-cpwd" role="alert"></span>
</div>

</fieldset>

<!-- ---- CAPTCHA (Advanced Extension) ---- -->
<fieldset>
    <legend>Security Check</legend>

    <p style="font-size:.85rem;color:var(--muted);margin-bottom:.8rem;">
        Type the characters shown in the image. Click the image to refresh.
    </p>

    <div class="captcha-wrap">
        <!-- Canvas CAPTCHA -->
        <canvas id="captcha-canvas" width="175" height="52"
            role="img"
            aria-label="CAPTCHA image – click to refresh"
            tabindex="0"></canvas>
        <button type="button" id="captcha-refresh"
            aria-label="Refresh CAPTCHA">🔄</button>
    </div>

    <div class="field-group" id="fg-captcha">
        <label for="captcha-input">
            Enter CAPTCHA <span class="required-star" aria-hidden="true">*</span>
        </label>
        <input type="text" id="captcha-input" name="captcha"
            required aria-required="true"
            aria-describedby="err-captcha"
            placeholder="Type 6 characters above"
            maxlength="6"
            autocomplete="off"
            style="text-transform:uppercase;letter-spacing:.2rem;" />
        <span class="val-icon" aria-hidden="true"></span>
        <span class="err-msg" id="err-captcha" role="alert"></span>
    </div>

```

```
</fieldset>
```

```
<!-- ---- Terms & Conditions ---- -->
```

```
<div class="field-group" id="fg-terms">
```

```
<label class="check-label">
```

```
<input type="checkbox" id="terms" name="terms"
```

```
  required aria-required="true"
```

```
  aria-describedby="err-terms" />
```

```
I have read and agree to the
```

```
<a href="#" target="_blank" rel="noopener">Terms & Conditions</a>
```

```
and <a href="#" target="_blank" rel="noopener">Privacy Policy</a>.
```

```
<span class="required-star" aria-hidden="true">*</span>
```

```
</label>
```

```
<span class="err-msg" id="err-terms" role="alert"></span>
```

```
</div>
```

```
<!-- ---- Buttons ---- -->
```

```
<button type="submit" class="submit-btn" id="submit-btn">
```

```
  Create My Account
```

```
</button>
```

```
<button type="reset" id="reset-btn"
```

```
style="display:block;width:100%;margin-top:.6rem;padding:.65rem;background:transparent;border:1.5px solid
```

```
var(--border);border-radius:var(--radius);cursor:pointer;font-size:.95rem;color:var(--muted);transition:border-color .2s,color .2s;"
```

```
onmouseover="this.style.borderColor='var(--danger)';this.style.color='var(--danger)'"
```

```
onmouseout="this.style.borderColor='var(--border)';this.style.color='var(--muted)'">
```

```
  Reset Form
```

```
</button>
```

```
</form>
```

```
<!-- ---- SUCCESS PANEL ---- -->
```

```
<div id="success-panel" tabindex="-1" aria-live="polite">
```

```
<div class="success-icon" aria-hidden="true">🎉</div>
```

```
<h2>Registration Successful!</h2>
```

```
<p>
```

```
  Welcome to StudentHub! Your account has been created.<br />
```

```
  Please check your email for a confirmation link.
```

```
</p>
```

```
<a href="../practical2/login.html"
  style="display:inline-block;margin-top:1.2rem;padding:.7rem
2rem;background:var(--primary);color:#fff;border-radius:var(--radius);text-decoration:no
ne;font-weight:700;">
  Go to Login
</a>
</div>
```

```
</div><!-- /form-card -->
```

```
<!-- =====
```

```
LIVE SUMMARY SIDEBAR
```

```
===== -->
```

```
<aside class="summary-card" aria-label="Registration summary" aria-live="polite">
```

```
<h2>📄 Form Summary</h2>
```

```
<div class="summary-row"><span class="s-label">First Name</span><span
class="s-val" id="s-fname">—</span></div>
```

```
<div class="summary-row"><span class="s-label">Last Name</span><span
class="s-val" id="s-lname">—</span></div>
```

```
<div class="summary-row"><span class="s-label">Email</span><span
class="s-val" id="s-email">—</span></div>
```

```
<div class="summary-row"><span class="s-label">Course</span><span
class="s-val" id="s-course">—</span></div>
```

```
<hr style="border-color:var(--border);margin:1rem 0;" />
```

```
<h2 style="font-size:.95rem;margin-bottom:.6rem;">🔑 Password
Requirements</h2>
```

```
<ul class="requirements-list" aria-label="Password strength requirements"
id="summary-req-list">
```

```
<li id="req-len-s" class="req-unmet"><span class="req-icon">○</span> 8+
characters</li>
```

```
<li id="req-upper-s" class="req-unmet"><span class="req-icon">○</span>
Uppercase letter</li>
```

```
<li id="req-lower-s" class="req-unmet"><span class="req-icon">○</span>
Lowercase letter</li>
```

```
<li id="req-num-s" class="req-unmet"><span class="req-icon">○</span>
Number</li>
```

```
<li id="req-special-s" class="req-unmet"><span class="req-icon">○</span> Special
character</li>
```

```
</ul>
```

```
<hr style="border-color:var(--border);margin:1rem 0;" />
```

```

    <p style="font-size:.78rem;color:var(--muted);">
      ✓ = Valid &nbsp; ✗ = Needs attention<br />
      Fields marked <span style="color:var(--danger);">*</span> are required.
    </p>
  </aside>

</div><!-- /page-wrap -->
</main>

<!-- FOOTER -->
<footer class="site-footer" role="contentinfo">
  <p>
    &copy; 2026 StudentHub – Practical 5: Registration Form Validation &nbsp;|&nbsp;
    <a href="../../practical2/index.html">← Home</a> &nbsp;|&nbsp;
    <a href="../../practical4/index.html">← Practical 4</a>
  </p>
</footer>

<script src="js/validate.js"></script>

<!-- Sync sidebar requirement list with main checklist -->
<script>
  // Mirror the password requirement states to the sidebar list
  const observer = new MutationObserver(() => {
    ['len','upper','lower','num','special'].forEach(key => {
      const src = document.getElementById('req-' + key);
      const dst = document.getElementById('req-' + key + '-s');
      if (!src || !dst) return;
      dst.className = src.className;
      const icon = dst.querySelector('.req-icon');
      const srclcon = src.querySelector('.req-icon');
      if (icon && srclcon) icon.textContent = srclcon.textContent;
    });
  });

  document.querySelectorAll('[id^="req-"]').forEach(el => {
    observer.observe(el, { attributes: true, attributeFilter: ['class'], childList: true, subtree:
true });
  });
</script>

</body>
</html>

```

Output:

StudentHub

Home Practical 4 Login

Create Your Account

Practical 5 – HTML5 inputs · JS Regex Validation · Real-time Feedback · Password Strength Meter · Canvas CAPTCHA

Already registered? [Login here.](#)

1. Personal Info

2. Academic

3. Security

Personal Information

First Name *

e.g. Rahul

Last Name *

e.g. Verma

Email Address *

Form Summary

First Name

Last Name

Email

Course

Password Requirements

☐ 8+ characters

☐ Uppercase letter

☐ Lowercase letter

☐ Number

☐ Special character

Email Address *

student@example.com

Mobile Number *

10-digit number

Date of Birth *

dd-mm-yyyy

Indian format: starts with 6-9

Gender *

☐ Male

☐ Female

☐ Other / Prefer not to say

Academic Information

Course *

-- Select Course --

Year of Study *

-- Select Year --

Roll Number (optional)

e.g. BCA/2024/021

Form Summary

First Name

Last Name

Email

Course

Password Requirements

☐ 8+ characters

☐ Uppercase letter

☐ Lowercase letter

☐ Number

☐ Special character

☒ = Valid ☐ = Needs attention

Password

- ☐ At least 8 characters
- ☐ At least one uppercase letter
- ☐ At least one lowercase letter
- ☐ At least one number
- ☐ At least one special character (@!%*?&_#^)

Re-enter your password

Type the characters shown in the image. Click the image to refresh.



First Name	—
Last Name	—
Email	—
Course	—

- ☐ 8+ characters
- ☐ Uppercase letter
- ☐ Lowercase letter
- ☐ Number
- ☐ Special character

✓ = Valid ✗ = Needs attention

Security Check

Type the characters shown in the image. Click the image to refresh.



TYPE 6 CHARACTERS ABOVE

☐ I have read and agree to the [Terms & Conditions](#) and [Privacy Policy](#). *

[Reset Form](#)

First Name _____

Last Name _____

Email _____

Course _____

- ☐ 8+ characters
- ☐ Uppercase letter
- ☐ Lowercase letter
- ☐ Number
- ☐ Special character

 = Valid = Needs attention
 Fields marked * are required.